

Dataset Description:

The dataset comprises time-series data collected by four sensors, which measure two target gases, Hydrogen Sulfide (H₂S) and Methyl Mercaptan (CH₃SH) in the presence of air. To obtain measurements, each gas was individually exposed to the multi-sensor setup, while maintaining room temperature. Table 2 in the description file presents the distribution of data samples for the target gases collected from the multi-sensor system against the true gas concentration in parts per million (ppm) at two different humidity levels. The dataset file is available in CSV format and contains 9 columns with a total of 654440x4 gas samples. The CSV file also includes additional information on temperature, humidity, and true concentrations of Hydrogen Sulfide and Methyl Mercaptan. Out of the 654440x4 samples, there are 151682x4 samples of Methyl Mercaptan, 126142x4 samples of Hydrogen Sulfide, and the remaining samples are normal air samples. This data set is used for gas classification using deep learning and data fusion to identify the target gases.

- No of Classes=3
- Class Names and corresponding labels= [0] for Air, [1] for Methyl Mercaptan (CH₃SH), [2] for Hydrogen Sulfide (H₂S)
- Number of sensors= 4
- Sampling rate= 1Hz
- Column names and order of CSV dataset file:

Table 1

Time[sec]	Sensor1[ppm]	Sensor2[ppm]	Sensor3[ppm]	Sensor4[ppm]	Humidity[%]	Temperature[C]	True_concentration[ppm]	Labels
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Table 2

True Gas concentration in (ppm)	Relative Humidity(%)	No of sample points representing multi-sensor response to target gas concentrations	
		CH ₃ SH	H ₂ S
2			
	30%	2240	2696
	50%	2375	2934
3			
	30%	4501	3401
	50%	2763	3726
4			
	30%	2782	2773
	50%	4479	2711
5			

	30%	3615	4582
	50%	8959	3742
6			
	30%	2815	4618
	50%	1893	1916
7			
	30%	2814	1904
	50%	3968	2799
8			
	30%	3924	4796
	50%	3751	2833
9			
	30%	3787	3919
	50%	5844	4795
10			
	30%	3811	3841
	50%	4683	4812
11			
	30%	4821	4868
	50%	4830	3877

12			
	30%	2885	4781
	50%	3056	4003
14			
	30%	2393	2907
	50%	2005	2000
15			
	30%	3960	1939
	50%	1855	1962
16			
	30%	2468	3394
	50%	2020	2017
18			
	30%	4623	3555
	50%	4171	2034
20			
	30%	6610	3714
	50%	3989	2050
22			
	30%	3937	4007

	50%	4310	3984
23			
	30%	4220	2025
	50%	4980	3984
25			
	30%	5990	3968
	50%	6066	2275
Total Samples		151682	126142